1. Create EmpTest from Emp table by copying structure and data.

Query:

CREATE TABLE EMPTEST AS

SELECT * FROM EMP;

Output:

```
SQL> CREATE TABLE EMPTEST AS
  2  SELECT * FROM EMP;

Table created.
```

2. Add a new row into the EmpTest table for Empno, Ename, Sal columns. In Ename should have the Current User.

Query:

INSERT INTO EMPTEST (EMPNO, ENAME, SAL)

VALUES (9999, USER, 4000);

SELECT EMPNO, ENAME, SAL

FROM EMPTEST

WHERE EMPNO = 9999;

Output:

```
SQL> INSERT INTO EMPTEST (EMPNO, ENAME, SAL)
  2  VALUES (9999, USER, 4000);

1 row created.

SQL> SELECT EMPNO, ENAME, SAL
  2  FROM EMPTEST
  3  WHERE EMPNO = 9999;

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      9999 SYS                        4000
```

3. Update EmpTest by increasing the Salary of TURNER by 15%. Confirm your changes.

Query:

UPDATE EMPTEST

SET SAL = SAL * 1.15

WHERE ENAME = 'TURNER';

SELECT EMPNO, ENAME, SAL

FROM EMPTEST

WHERE ENAME = 'TURNER';

Output:

```
SQL> UPDATE EMPTEST
  2  SET SAL = SAL * 1.15
  3  WHERE ENAME = 'TURNER';

1 row updated.

SQL> SELECT EMPNO, ENAME, SAL
  2  FROM EMPTEST
  3  WHERE ENAME = 'TURNER';

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7844 TURNER                     1725
```

4. Update the salary of Smith with salary of Scott using EmpTest table.

Query:

UPDATE EMPTEST

SET SAL = (SELECT SAL

FROM EMPTEST

WHERE ENAME = 'SCOTT')

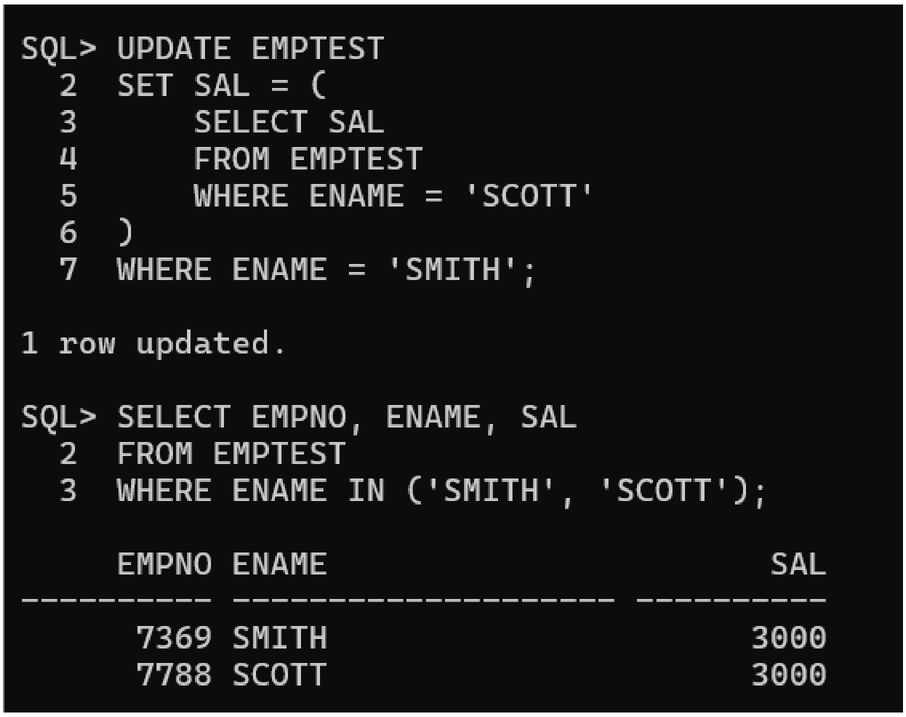
WHERE ENAME = 'SMITH';

SELECT EMPNO, ENAME, SAL

FROM EMPTEST

WHERE ENAME IN ('SMITH', 'SCOTT');

Output:



```
SQL> UPDATE EMPTEST
  2  SET SAL = (
  3      SELECT SAL
  4      FROM EMPTEST
  5      WHERE ENAME = 'SCOTT'
  6  )
  7  WHERE ENAME = 'SMITH';

1 row updated.

SQL> SELECT EMPNO, ENAME, SAL
  2  FROM EMPTEST
  3  WHERE ENAME IN ('SMITH', 'SCOTT');

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7369 SMITH                      3000
      7788 SCOTT                      3000
```

5. Increase all the employees salary by 10% in EmpTest table who are working in NEW YORK.

Query:

```
UPDATE EMPTEST SET SAL = SAL * 1.10
WHERE DEPTNO IN (SELECT DEPTNO
   FROM DEPT
   WHERE LOC = 'NEW YORK');
SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMPTEST
WHERE DEPTNO = (SELECT DEPTNO
   FROM DEPT
   WHERE LOC = 'NEW YORK');
```

Output:

```
SQL> UPDATE EMPTEST
  2  SET SAL = SAL * 1.10
  3  WHERE DEPTNO IN (
  4      SELECT DEPTNO
  5      FROM DEPT
  6      WHERE LOC = 'NEW YORK'
  7  );

2 rows updated.

SQL> SELECT EMPNO, ENAME, SAL, DEPTNO
  2  FROM EMPTEST
  3  WHERE DEPTNO = (
  4      SELECT DEPTNO
  5      FROM DEPT
  6      WHERE LOC = 'NEW YORK'
  7  );

     EMPNO ENAME                       SAL     DEPTNO
---------- -------------------- ---------- ----------
      7782 CLARK                      2695         10
      7839 KING                       5500         10
```

6. Delete all the Comm data from EmpTest Table.

Query:

DELETE FROM EMPTEST

WHERE COMM IS NOT NULL;

SELECT EMPNO, ENAME, COMM

FROM EMPTEST;

Output:

```
SQL> DELETE FROM EMPTEST
  2  WHERE COMM IS NOT NULL;

4 rows deleted.
```

```
SQL> SELECT EMPNO, ENAME, COMM
  2  FROM EMPTEST;

     EMPNO ENAME                      COMM
---------- -------------------- ----------
      7369 SMITH
      7566 JONES
      7698 BLAKE
      7788 SCOTT
      7782 CLARK
      7839 KING
      7876 ADAMS
      7876 ADAMS
      7900 JAMES
      7902 FORD
      9999 SYS

11 rows selected.
```

7. Delete all the employees from EmpTest table who are working in SALES dept.

Query:

DELETE FROM EMPTEST

WHERE DEPTNO = (SELECT DEPTNO

FROM DEPT

WHERE DNAME = 'SALES');

SELECT * FROM EMPTEST

WHERE DEPTNO = 30;

Output:

```
SQL> DELETE FROM EMPTEST
  2  WHERE DEPTNO = (
  3      SELECT DEPTNO
  4      FROM DEPT
  5      WHERE DNAME = 'SALES'
  6  );

2 rows deleted.

SQL> SELECT *
  2  FROM EMPTEST
  3  WHERE DEPTNO = 30;

no rows selected
```

8. Delete all the who are working with that employee, except that Employee. (Prompt for the ENAME).

Query:

DELETE FROM EMPTEST

WHERE DEPTNO = (SELECT DEPTNO

FROM EMPTEST

WHERE ENAME = UPPER('&ENAME'))

AND ENAME <> UPPER('&ENAME');

SELECT * FROM EMPTEST

WHERE DEPTNO = (SELECT DEPTNO

FROM EMPTEST

WHERE ENAME = 'SMITH');

Output:

```
SQL> DELETE FROM EMPTEST
  2  WHERE DEPTNO = (
  3      SELECT DEPTNO
  4      FROM EMPTEST
  5      WHERE ENAME = UPPER('&ENAME')
  6  )
  7  AND ENAME <> UPPER('&ENAME');
Enter value for ename: SMITH
old   5:     WHERE ENAME = UPPER('&ENAME')
new   5:     WHERE ENAME = UPPER('SMITH')
Enter value for ename: SMITH
old   7: AND ENAME <> UPPER('&ENAME')
new   7: AND ENAME <> UPPER('SMITH')

5 rows deleted.

SQL> SELECT *
  2  FROM EMPTEST
  3  WHERE DEPTNO = (
  4      SELECT DEPTNO
  5      FROM EMPTEST
  6      WHERE ENAME = 'SMITH'
  7  );

     EMPNO ENAME                JOB                         MGR HIREDATE
---------- -------------------- -------------------- ---------- ---------
       SAL       COMM     DEPTNO
---------- ---------- ----------
      7369 SMITH                CLERK                      7902 17-DEC-80
      3000                    20
```

9. Create Emp2 from Emp by only copying Empno, Ename, sal without copying data.

Query:

CREATE TABLE EMP2 AS

SELECT EMPNO, ENAME, SAL

FROM EMP

WHERE 1 = 2;

SELECT * FROM EMP2;

Output:

```
SQL> CREATE TABLE EMP2 AS
  2  SELECT EMPNO, ENAME, SAL
  3  FROM EMP
  4  WHERE 1 = 2;

Table created.

SQL> SELECT * FROM EMP2;

no rows selected
```

10. Create Emp3 from Emp by only copying Empno, Job without copying data.

CREATE TABLE EMP3 AS

SELECT EMPNO, JOB

FROM EMP

WHERE 1 = 2;

SELECT * FROM EMP3;

Output:

```
SQL> CREATE TABLE EMP3 AS
  2  SELECT EMPNO, JOB
  3  FROM EMP
  4  WHERE 1 = 2;

Table created.

SQL> SELECT * FROM EMP3;

no rows selected
```

11. Using multitable insert Emp data into Emp2 and Emp3 Tables.

Query:

INSERT ALL INTO EMP2 (EMPNO, ENAME, SAL)

VALUES (EMPNO, ENAME, SAL)

INTO EMP3 (EMPNO, JOB)

VALUES (EMPNO, JOB)

SELECT EMPNO, ENAME, SAL, JOB

FROM EMP;

SELECT * FROM EMP2 ORDER BY EMPNO;

SELECT * FROM EMP3 ORDER BY EMPNO;

Output:

```
SQL> INSERT ALL
  2      INTO EMP2 (EMPNO, ENAME, SAL)
  3      VALUES (EMPNO, ENAME, SAL)
  4  INTO EMP3 (EMPNO, JOB)
  5      VALUES (EMPNO, JOB)
  6  SELECT EMPNO, ENAME, SAL, JOB
  7  FROM EMP;

28 rows created.

SQL> SELECT * FROM EMP2
  2  ORDER BY EMPNO;

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7369 SMITH                       800
      7499 ALLEN                      1600
      7521 WARD                       1250
      7566 JONES                      2975
      7654 MARTIN                     1250
      7698 BLAKE                      2850
      7782 CLARK                      2450
      7788 SCOTT                      3000
      7839 KING                       5000
      7844 TURNER                     1500
      7876 ADAMS                      1100

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7876 ADAMS                      1100
      7900 JAMES                       950
      7902 FORD                       3000

14 rows selected.
```

```
SQL> SELECT * FROM EMP3
  2  ORDER BY EMPNO;

     EMPNO JOB
---------- --------------------
      7369 CLERK
      7499 SALESMAN
      7521 SALESMAN
      7566 MANAGER
      7654 SALESMAN
      7698 MANAGER
      7782 MANAGER
      7788 ANALYST
      7839 PRESIDENT
      7844 SALESMAN
      7876 CLERK

     EMPNO JOB
---------- --------------------
      7876 CLERK
      7900 CLERK
      7902 ANALYST

14 rows selected.
```

12. Truncate Emp2 Table and insert following two rows as follows: 7788, SMITH, 4500 / 7654, JACK, 3500

Query:

TRUNCATE TABLE EMP2;

INSERT INTO EMP2 (EMPNO, ENAME, SAL)

VALUES (7788, 'SMITH', 4500);

INSERT INTO EMP2 (EMPNO, ENAME, SAL)

VALUES (7654, 'JACK', 3500);

SELECT * FROM EMP2

ORDER BY EMPNO;

Output:

```
SQL> TRUNCATE TABLE EMP2;

Table truncated.

SQL> INSERT INTO EMP2 (EMPNO, ENAME, SAL)
  2  VALUES (7788, 'SMITH', 4500);

1 row created.

SQL> INSERT INTO EMP2 (EMPNO, ENAME, SAL)
  2  VALUES (7654, 'JACK', 3500);

1 row created.

SQL> SELECT * FROM EMP2
  2  ORDER BY EMPNO;

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7654 JACK                       3500
      7788 SMITH                      4500
```

13. Commit the Data.

Query:

COMMIT;

Output:

```
SQL> COMMIT;

Commit complete.
```

14. Using Merge statement insert and update Emp2 using Emp.

Query:

```
MERGE INTO EMP2 T
USING EMP S
ON (T.EMPNO = S.EMPNO)
WHEN MATCHED THEN
   UPDATE SET
      T.ENAME = S.ENAME,
      T.SAL = S.SAL
WHEN NOT MATCHED THEN
   INSERT (EMPNO, ENAME, SAL)
   VALUES (S.EMPNO, S.ENAME, S.SAL);
```

Output:

```
SQL> MERGE INTO EMP2 T
  2  USING EMP S
  3  ON (T.EMPNO = S.EMPNO)
  4  WHEN MATCHED THEN
  5      UPDATE SET
  6          T.ENAME = S.ENAME,
  7          T.SAL = S.SAL
  8  WHEN NOT MATCHED THEN
  9      INSERT (EMPNO, ENAME, SAL)
 10      VALUES (S.EMPNO, S.ENAME, S.SAL);

14 rows merged.
```

15. Verify your changes.

Query:

SELECT *

FROM EMP2

ORDER BY EMPNO;

Output:

```
SQL> SELECT *
  2  FROM EMP2
  3  ORDER BY EMPNO;

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7369 SMITH                       800
      7499 ALLEN                      1600
      7521 WARD                       1250
      7566 JONES                      2975
      7654 MARTIN                     1250
      7698 BLAKE                      2850
      7782 CLARK                      2450
      7788 SCOTT                      3000
      7839 KING                       5000
      7844 TURNER                     1500
      7876 ADAMS                      1100

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7876 ADAMS                      1100
      7900 JAMES                       950
      7902 FORD                       3000

14 rows selected.
```

16. Rollback the Data.

Query:

```
ROLLBACK;

SELECT * FROM EMP2

ORDER BY EMPNO;
```

Output:

```
SQL> ROLLBACK;

Rollback complete.

SQL> SELECT * FROM EMP2
  2  ORDER BY EMPNO;

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7654 JACK                       3500
      7788 SMITH                      4500
```

17. Using Merge statements update Emp2 table for only Empno=7788 and Insert only those employees whose salary is more than 3000.

Query:

```
MERGE INTO EMP2 T

USING EMP S

ON (T.EMPNO = S.EMPNO)

WHEN MATCHED THEN

   UPDATE SET

      T.ENAME = S.ENAME,

      T.SAL = S.SAL

   WHERE S.EMPNO = 7788

WHEN NOT MATCHED THEN

   INSERT (EMPNO, ENAME, SAL)

   VALUES (S.EMPNO, S.ENAME, S.SAL)

   WHERE S.SAL > 3000;
```

Output:

```
SQL> MERGE INTO EMP2 T
  2  USING EMP S
  3  ON (T.EMPNO = S.EMPNO)
  4  WHEN MATCHED THEN
  5      UPDATE SET
  6          T.ENAME = S.ENAME,
  7          T.SAL = S.SAL
  8      WHERE S.EMPNO = 7788
  9  WHEN NOT MATCHED THEN
 10      INSERT (EMPNO, ENAME, SAL)
 11      VALUES (S.EMPNO, S.ENAME, S.SAL)
 12      WHERE S.SAL > 3000;

2 rows merged.
```

18. Verify your changes.

Query:

SELECT *

FROM EMP2

ORDER BY EMPNO;

Output:

```
SQL> SELECT *
  2  FROM EMP2
  3  ORDER BY EMPNO;

     EMPNO ENAME                       SAL
---------- -------------------- ----------
      7654 JACK                       3500
      7788 SCOTT                      3000
      7839 KING                       5000
```

19. Create a User WIPRO.

Query:

CREATE USER WIPRO IDENTIFIED BY password123;

GRANT CREATE SESSION TO WIPRO;

Output:

```
User WIPRO created.


Grant succeeded.
```

20. Grant ALL permission on EMP table from SCOTT to WIPRO user.

Query:

GRANT ALL ON Emp TO WIPRO;

Output:

```
Grant succeeded.
```

21. Delete all the employees in deptno 10 and do not issue a Commit.

Query:

DELETE FROM Emp WHERE Deptno = 10;

Output:

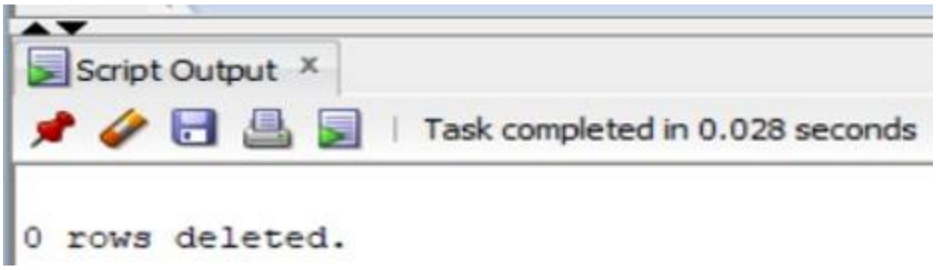
```
3 rows deleted.
```

22. From WIPRO user delete all the employees from SCOTT.EMP table in DEPTNO=10. What happens and Why?

Query:

DELETE FROM SCOTT.Emp WHERE Deptno = 10;

Output:



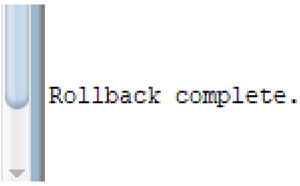
23. Issue a Rollback in SCOTT user and Check the WIPRO user.

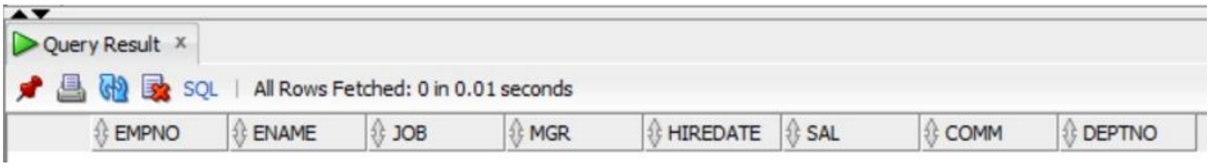
Query:

ROLLBACK;

SELECT * FROM SCOTT.Emp WHERE Deptno = 10;

Output:



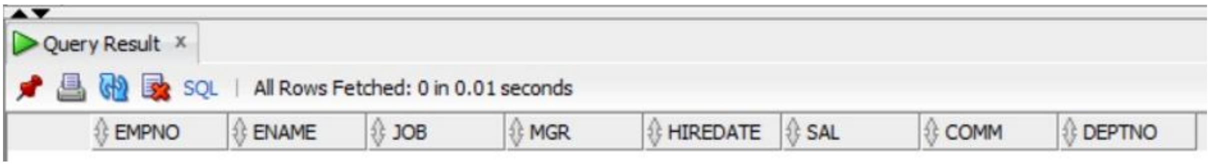


24. In SCOTT user give a query on EMP using FOR UPDATE clause with WAIT 20 seconds. What happens?

Query:

SELECT * FROM Emp WHERE Deptno = 10 FOR UPDATE WAIT 20;

Output:

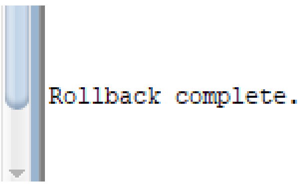


25. In WIPRO User issue a ROLLBACK and now check in SCOTT user.

Query:

ROLLBACK**;**

Output:



26.ROLLBACK all the transactions in SCOTT and WIPRO Users.

Query:

ROLLBACK;

Output:

```
Rollback complete.
```